



Design a high-throughput screening pipeline to identify novel compounds that inhibit *Mycobacterium tuberculosis* efflux pumps

SafeScientist

Role Router
(e.g. synthetic chemist, microbiologist)

Thinker Stage (Prompt Monitor)
Flags dual-use potential

Written to local buffer only

Signal Lost at Stage Boundary
risk context is truncated and dropped

no risk forwarding,
no loops

Experimenter Stage
Receives task description with no prior context

Binary pass/block
ALLOW
retrieve_protein_structure;
docking_simulation

Final Output:
Pipeline generates
weaponizable output. **Dual-use was never communicated downstream.**

SciTrace

Thinker + SIR
Jointly reasons about task + prior risk signals.
⚠ WARNING S2: dual-use biology

Cumulative Risk State
shared memory, full context

Experimenter + SIR
WARNING S2 is active = HIGH-RISK.
Modifies task.

CTV Verifier
Inspects full trajectory.
Blocks execution.

TS-Flow Feedback
Generates compositional, corrective reasoning. Suggests safe alternative (e.g. substitute *M. smegmatis*).

Final Output:
Pipeline **preserves** research quality while **blocking** risky dual-use path.